ASSIGNMENT 1: ADR IDENTIFICATION BASICS

CASE ANALYSIS: LAMOTRIGINE-ASSOCIATED STEVENS–JOHNSON SYNDROME

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1. PATIENT INFORMATION

- Age: 29 years
- Gender: Female
- Medical history: Recurrent depressive disorder
- Hospital: KIST Medical College and Teaching Hospital
- Date of ICU admission: 1 April 2024

2. SUSPECTED ADVERSE DRUG REACTION

Suspected ADR: Stevens–Johnson syndrome (SJS)/toxic epidermal necrolysis (TEN) overlap.

Clinical manifestations

The patient presented with:

- Generalized maculopapular rash with itching and painful raised lesions.
- Rash spreading from the upper trunk to the extremities, palms, soles and face.
- Oral mucosal erosion.
- Difficulty eating and swallowing.
- Itching of the eyes.
- Fever, malaise and sore throat.
- Facial edema, particularly around the eyes and lips.

The patient was diagnosed with SJS/TEN and admitted to the ICU for further management.

3. SUSPECTED MEDICATION HISTORY

The following medications were prescribed for recurrent depressive disorder:

Suspected/concomitant Medication	Dose	Frequency	Start Date
Lamotrigine	50 mg	Twice daily	12/03/2024
Olanzapine	5 mg	Once daily	12/03/2024
Sertraline	100 mg	Once daily	12/03/2024

On 19 March 2024, the dose of lamotrigine was increased to 100 mg twice daily.

All three medications were withheld following the patient's ICU admission.

4. TIMELINE ANALYSIS

Date	Event
12/03/2024	Lamotrigine, olanzapine and sertraline initiated.
19/03/2024	Lamotrigine dose increased to 100 mg twice daily.
Approximately one week after dose escalation	Patient developed skin rash and associated symptoms.
1/04/2026	Patient admitted to ICU with generalized rash and mucosal involvement.
Following admission	All previous medications withheld; treatment initiated.

The onset of symptoms after lamotrigine initiation and dose escalation indicates a temporal relationship between drug exposure and the suspected ADR.

5. IDENTIFICATION OF THE SUSPECTED DRUG

Primary suspected drug: **Lamotrigine**

Lamotrigine is associated with serious cutaneous adverse reactions, including Stevens–Johnson syndrome and toxic epidermal necrolysis.

The following findings support lamotrigine as the primary suspected drug:

- The reaction occurred after initiation of lamotrigine.
- Symptoms developed after an increase in the lamotrigine dose.
- Lamotrigine is a recognized cause of SJS/TEN.
- The patient's clinical manifestations were consistent with SJS/TEN.

Olanzapine and sertraline were also administered. Therefore, their contribution cannot be completely excluded. However, lamotrigine was considered the more likely causative medication in the published case report.

6. CONCLUSION

The patient developed Stevens–Johnson syndrome/toxic epidermal necrolysis overlap following the initiation and dose escalation of lamotrigine.

The temporal relationship, clinical manifestations and known association between lamotrigine and serious cutaneous adverse reactions support the identification of lamotrigine as the primary suspected drug.

7. PUBLISHED CASE REFERENCE

Title: *A Singular Case Analysis: Lamotrigine-Associated Stevens–Johnson Syndrome*
Authors: Albin Joshi, Anjula Palikhe, Santosh Acharya and Puskar Kunwor.
Journal: Case Reports in Critical Care. 2024

Published case link:
https://pmc.ncbi.nlm.nih.gov/articles/PMC11614506/

NOTE : This assignment is based on the published clinical case report cited above.